LINEAR ALGEBRA · LESSON 9

Independence, Basis and Dimension

Which funds are **genuinely different**? What's the smallest menu that still reaches everything?

How many independent bets does a menu really offer?

Basis | n = r + (n – r) | MIT 18.06, Lecture 9

No fund can be rebuilt from the others

$c_1 v_1 + \cdots + c_n v_n = 0$ only when every $c = 0$.

Dependent example:

$$-\tfrac{1}{2}\begin{bmatrix}1\\12\\1\end{bmatrix} - \tfrac{1}{2}\begin{bmatrix}1\\7\\0\end{bmatrix} + \begin{bmatrix}1\\9.5\\0.5\end{bmatrix} = 0$$

Balanced $= \tfrac{1}{2}$ Nifty $+ \tfrac{1}{2}$ bonds. Dependent = a do-nothing trade exists.

THE TEST

Put them in columns, count pivots

Columns independent $\Leftrightarrow$ $N(A) = \{0\}$ $\Leftrightarrow$ no free variables $\Leftrightarrow$ **rank = n.**

Funds	Pivots	Independent?
Nifty, bonds, midcap	3 of 3	yes
Nifty, bonds, balanced	2 of 3	no

More vectors than entries ($n > m$) $\Rightarrow$ always dependent.

Reach everything, waste nothing

Span: all combinations = every reachable target.

Basis: independent **and** spanning: the smallest complete menu.

> **STANDARD BASIS = ARROW–DEBREU**
> $(1,0,0), (0,1,0), (0,0,1)$: each pays 1 in exactly one scenario.

Nifty, bonds, midcap are another basis of $\mathbb{R}^3$.

DIMENSION

Every basis has the same size

That number is the **dimension**: the number of genuinely different directions.

- $\mathbb{R}^3$: dimension 3
- The plane $R = 7B + 5E$: dimension 2

Matrices have a rank; spaces have a dimension: $\text{rank}(A) = \dim C(A)$.

Pivot columns and special solutions

Basis of $C(A)$: the pivot columns **of A.** $\dim = r$.

Basis of $N(A)$: the special solutions. $\dim = n - r$.

> **CAREFUL**
> Use A's columns, not R's: row operations change the column space.

FIVE FUNDS

Only 3 real directions

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 & 0 \\ 0 & 0 & \boxed{1} & 0.5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Nifty, ETF, bonds, balanced, midcap → pivots: Nifty, bonds, midcap.

$\dim C(A) = 3$ (every target), $\dim N(A) = 2$ (ETF = Nifty; balanced = mix).

THE COUNT

n = r + (n – r)

Every fund is either a **new direction** (pivot) or a **redundancy** (free).

$$5 \text{ funds} = 3 \text{ directions} + 2 \text{ redundancies}$$

> AS A TRADER
>
> 50 products driven by 3 risk factors: at most 3 independent bets, at least 47 redundant routes.

IN MARKETS

A basis of payoffs

Futures, 20,000 call, 20,000 put, 19,900 call over 3 scenarios:

- Basis of $C(A)$: futures, 20,000 call, 19,900 call (dim 3)
- Basis of $N(A)$: $(1, -1, 1, 0)$, put–call parity (dim 1)

The put is redundant given futures and the call.

RECAP

Key takeaways

1. Independent $\Leftrightarrow$ rank $= n \Leftrightarrow N(A) = \{0\}$.
2. Basis = independent + spans.
3. Dimension = vectors in any basis.
4. Basis of $C(A)$: pivot columns of A.
5. Basis of $N(A)$: special solutions.
6. $n = r + (n - r)$.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A fund picker, a basis and dimension lab for any vectors, and worked hedging and options examples:

learn.krisarj.com

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